SABRE: An Uncertainty-Aware Modular Semantic–Behavioural Triage Framework for Adaptive Ransomware Detection

Henry Kabuye · Biju Issac · Neera Jeyamohan
Northumbria University

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Ransomware Defence Is a Decision Problem

Current detectors give a binary answer — but deployment demands more

- False automated containment = operational disruption (FP cost)
- Missed detection = encryption, data loss, ransom payment (FN cost)
- Novel families and obfuscated variants degrade confidence silently
- Analyst workload spikes when every alert demands manual review

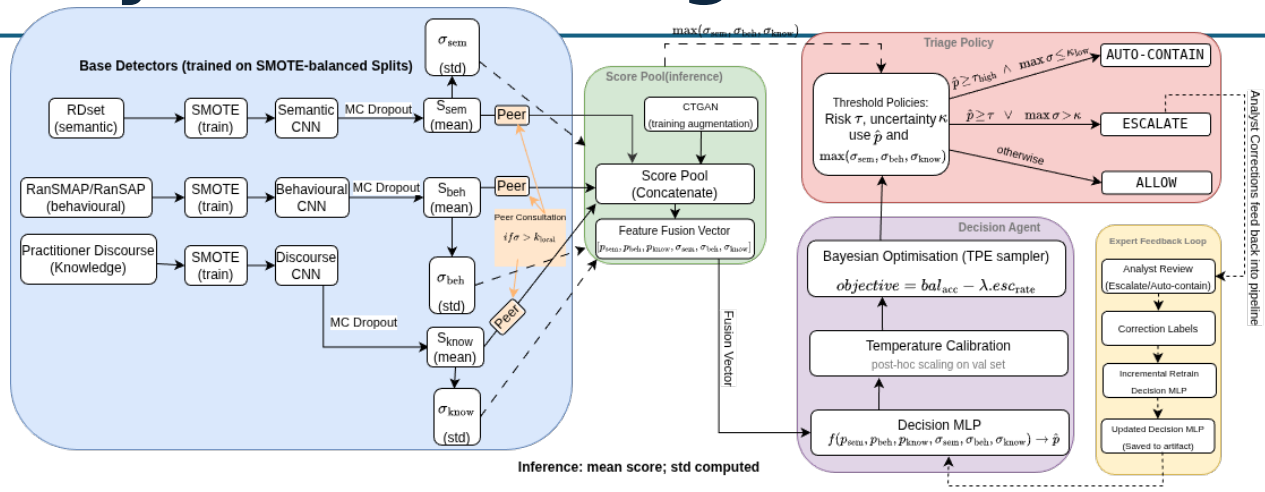
What is missing from the state of practice:

- Principled uncertainty — how confident is the detector, and why?
- Actionable triage — automated on confident cases, escalated on uncertain ones
- Modality-aware fusion — semantic PE signals ≠ runtime telemetry signals

SABRE operationalises uncertainty as a control variable for safer deployment

SABRE: Two Complementary Detection Agents

- **Semantic Agent**
  - Input: PE file → section names → sentence transformer (384D)
  - Detects structural packing and obfuscation patterns
- **Behavioural Agent**
  - Input: 10-second telemetry windows → 16D features
  - Storage: LBA variance, write entropy, throughput
  - Memory: GPA variance, page counts, entropy
- **Uncertainty Quantification**
  - Monte Carlo Dropout (T=10 passes) per agent
  - Each agent outputs: prediction p̂ + uncertainty σ
- **Score fusion → triage policy → AUTO-CONTAIN / ESCALATE / ALLOW**

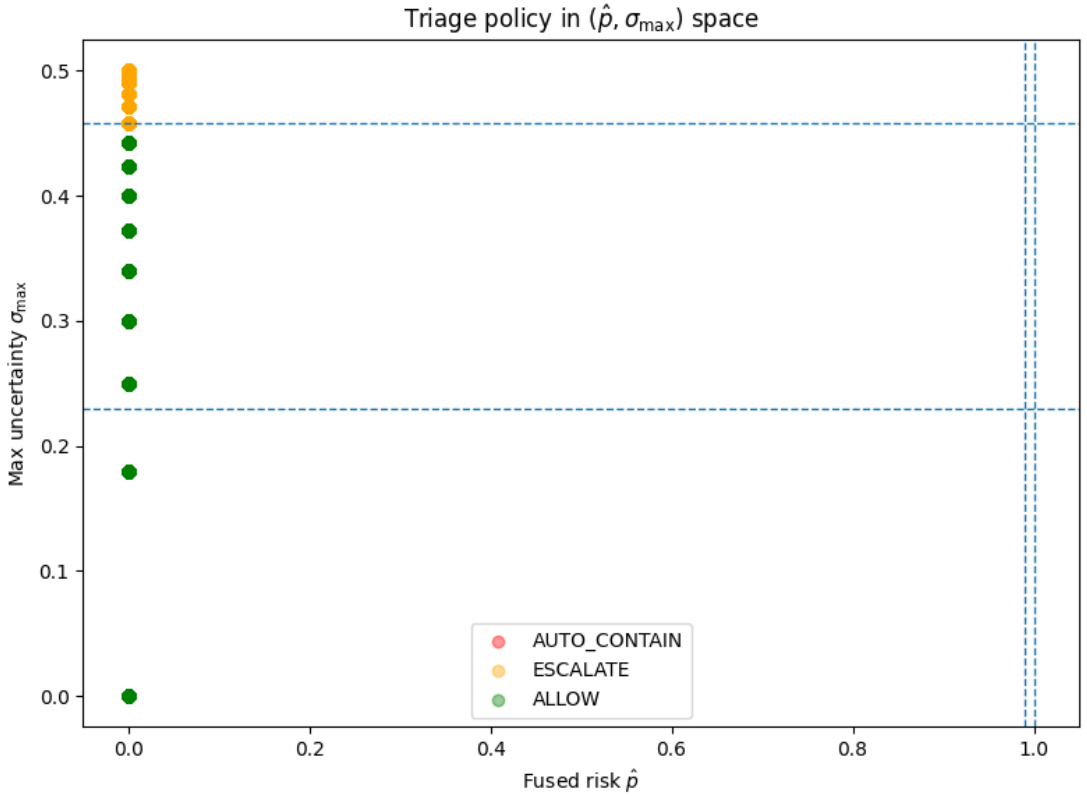


Operationalising Uncertainty: The Triage Policy

Three-way decision (not binary):

- **AUTO-CONTAIN**
  - $\hat{p} \geq \tau_high$ AND $\sigma_max \leq \kappa_low$
  - High confidence ransomware → act immediately
- **ESCALATE**
  - $\hat{p} \geq \tau$ AND $\sigma_max > \kappa$ (uncertain but suspicious)
  - OR $\hat{p}$ high but evidence conflicts across agents
  - → Route to analyst; uncertainty is the trigger
- **ALLOW**
  - Confident benign prediction → pass through

Key insight: uncertainty gates automation — the system knows when NOT to act autonomously



Results: Where SABRE Works and Where It Struggles

Semantic setting (RDset — 138K PE files):

- Semantic agent: AUROC 0.996 — strong PE separation
- PE structural features discriminate clearly

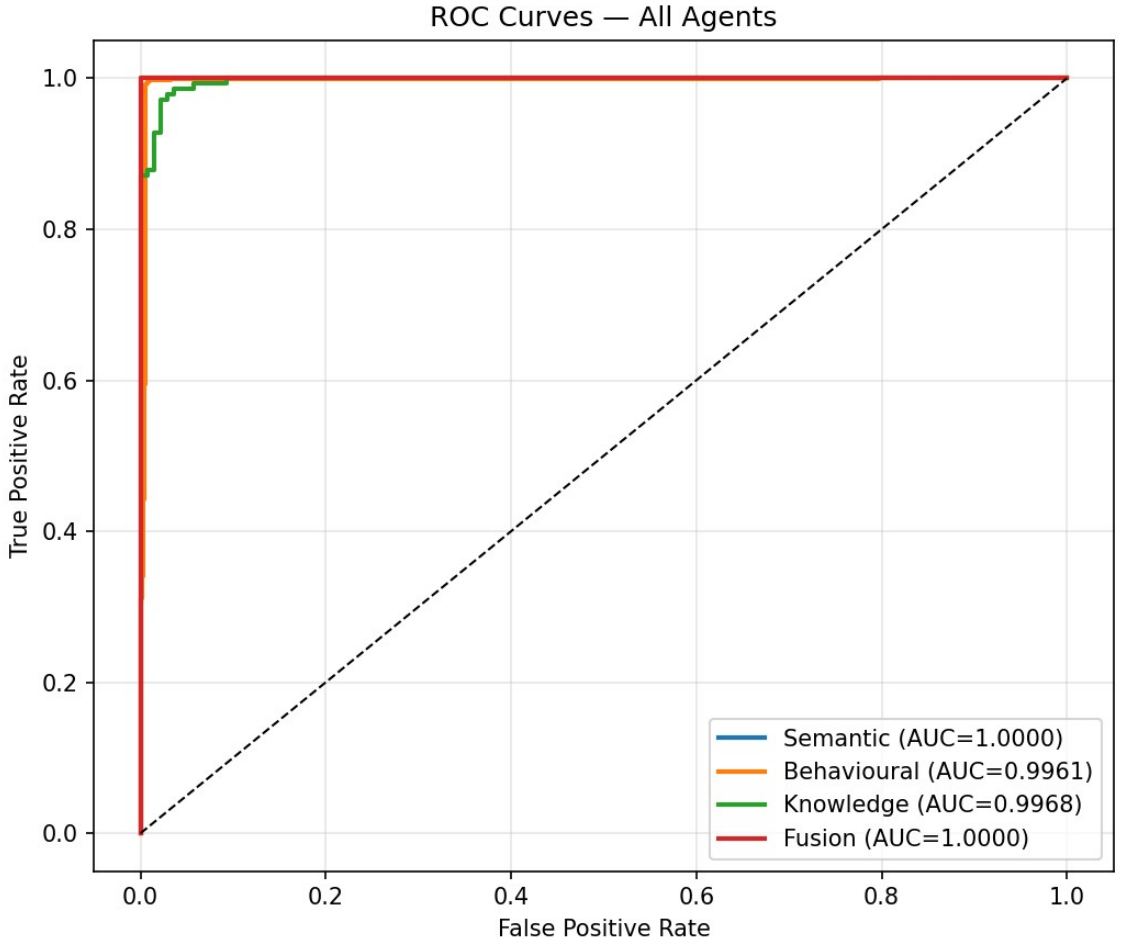
Behavioural setting (RanSMAP — 35,857 windows, 15 hardware configs):

- Behavioural agent: AUROC 0.963 (random split)
- Family holdout: AUROC drops to 0.706 for unseen families
- Concurrent execution (AESCrypt + REvil): detection rate 1.5%

Why does this matter?

- Weaker behavioural evidence → σ_beh rises → ESCALATE fires
- Uncertainty correctly identifies where the model is least reliable
- Novel families not seen at training → higher σ → analyst review

The gap between settings is not a failure — it is informative



Explainability: Making Decisions Auditable

Behavioural — Permutation Feature Importance:

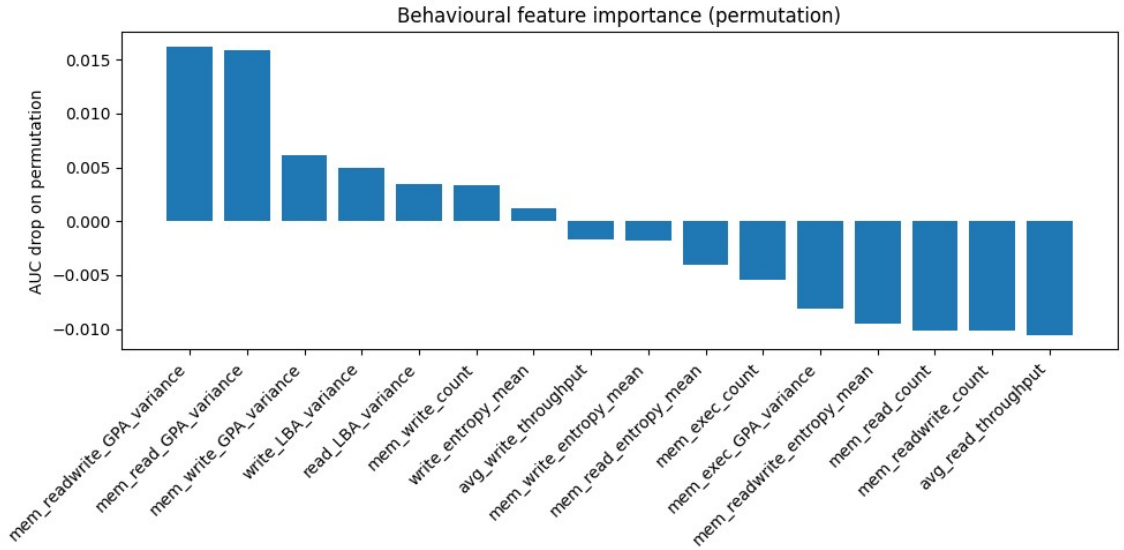
- write_entropy_mean: top driver — ransomware encrypts → max entropy
- write_LBA_variance: sequential writes across entire disk
- mem_write_GPA_variance: broad memory pages written uniformly

Semantic — Counterfactual Diagnostics:

- For borderline PE files: minimal embedding shift to flip verdict
- Short L_2 distance → closer to decision boundary → escalate
- Supports audit: 'why was this file flagged?'

On Reviewer 3's question — current operational practice:

- Antivirus/EDR: threshold-based scores, no principled uncertainty
- SABRE adds: σ as a first-class decision variable + XAI audit trail
- Analyst can see WHY a sample was escalated, not just THAT it was



Since This Abstract: SABRE-II — Learned Fusion & Policy

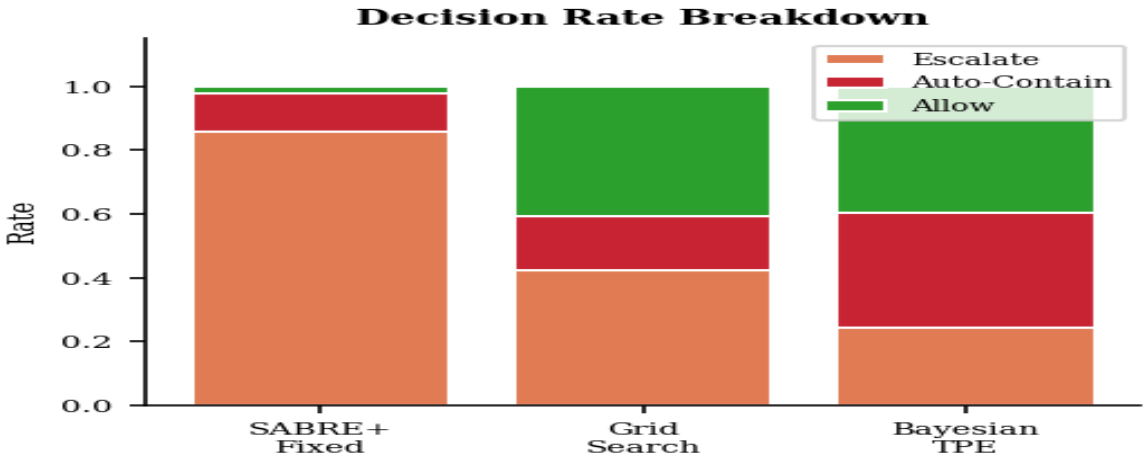
SABRE (this abstract)

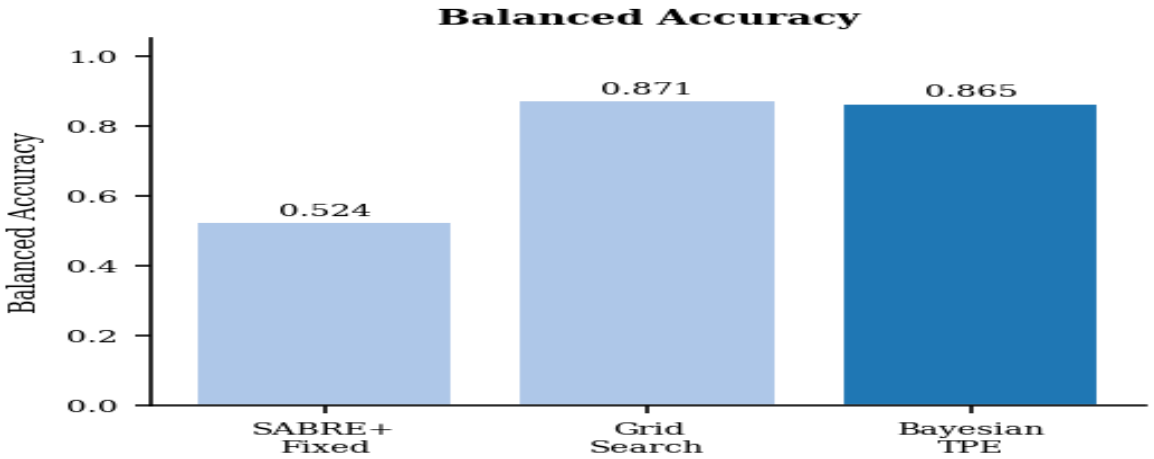
- Mean score fusion (equal weights)
- Fixed hand-tuned triage thresholds τ, κ
- 2 agents: Semantic + Behavioural
- MC Dropout uncertainty per agent
- AUROC 0.963 (random), 0.706 (family holdout)

→

SABRE-II (current work)

- Decision MLP replaces mean fusion
  - ECE: 0.094 → 0.018 (calibration gain)
- CTGAN score-space augmentation
- Bayesian TPE policy optimisation (300 trials)
  - Balanced acc: 0.525 → 0.866
  - Escalation rate: 85% → 25% (Bayesian vs fixed)
- Peer Consultation: prevents knowledge collapse
- AUROC 0.975 (random), 0.980 (temporal)





Conclusions

SABRE makes uncertainty operational — not merely descriptive

- Triage converts $(\hat{p}, \sigma) \rightarrow$ actionable decision: automate vs escalate vs allow
- Uncertainty correctly identifies hardest cases: novel families, concurrent execution
- XAI (feature importance, counterfactuals) supports analyst audit
- SABRE-II: learned fusion + Bayesian policy reduces unnecessary escalation by 70%

Publications:

- ESWA (Expert Systems With Applications) — under review, 2nd revision
- IEEE TIFS — SABRE+ submitted; SABRE-II in preparation

Questions welcome on:

- Triage threshold design and cost-sensitivity
- Generalisation to unseen ransomware families (family holdout results)
- Comparison with operational EDR/AV uncertainty practices